

Harim Choi

MACHINE LEARNING ENGINEER | PREDICTION · UNCERTAINTY · PRODUCTION ML

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Korean (native), English (professional working proficiency)

SUMMARY

I am a machine learning engineer interested in vectorizing reality, making predictions, and estimating uncertainty. Over seven years, I have worked with public-procurement, construction, and retail data, connecting tabular ML, NLP, computer vision, and LLM agents from data collection through evaluation and deployment. I quickly learn unfamiliar domains and inherited systems, then turn data, models, and operating workflows into working services. I build systems that balance predictive performance with reliability, cost, and operational constraints.

CORE CAPABILITIES

- Rapid Problem Understanding & Execution | Quickly learn unfamiliar domains and inherited systems, prioritize the highest-impact problems, and turn them into operable ML systems.
- Model Diagnosis & Improvement | Identify implementation errors and performance bottlenecks through experiments and analysis, then improve training, inference, and evaluation for the operating context.
- Uncertainty-Aware Decision Support | Go beyond point predictions by using distributions, intervals, calibration, and simulation to communicate reliability and risk.
- Research-to-Production | Adapt new research methods to real data, infrastructure, cost, and operating constraints, then implement them as repeatable pipelines.
- End-to-End ML Delivery | Connect data construction, training, evaluation, batch jobs, APIs, and service operations in a form that others can review, maintain, and extend.

TECHNICAL STACK

- Tabular & Probabilistic ML | TabPFN · TabICLv2 · Prior-Data Fitted Networks · Tabular ICL · AutoGluon · Quantile/Distributional Prediction · Conformal Prediction · Calibration · Monte Carlo Simulation
- NLP & Document Intelligence | Transformers · PEFT/LoRA · Weak Supervision · Knowledge Distillation · OCR · Document Parsing · Information Extraction
- Computer Vision | Recognition · Object Detection & Tracking · Segmentation · 2D/3D Pose · Motion & Trajectory Prediction
- RAG & Agentic Systems | Vector Retrieval · Hybrid Search · Reranking · LangGraph · MCP · Tool Calling · HITL
- Efficient Model Inference | INT3/INT8 Quantization · Weight/KV/Activation Compression · ONNX CPU/GPU Inference
- Production ML Systems | Data/Training Pipelines · Parallel Inference · API/Batch Serving · Model/Artifact Lifecycle · Monitoring

EXPERIENCE

Sejong Bid Institute | Researcher

Apr 2026–Present

Public-procurement probabilistic ML, Document AI, and production operations

- **Probabilistic Bid ML & MLOps** | Took over a public-procurement AI system built by a five-person team over six months and rebuilt it in three months. Modeled regional competitor bids with quantile models and CQR/R2CCP, then generated about 80,000 candidate bids per notice from a q1000 model zoo using TabICLv2, CatBoost, XGBoost, and AutoGluon. Connected candidates to Monte Carlo simulation and operated the workflow through Airflow DAGs, parallel GPU inference, artifact checksums, dry-runs, and approval-gated deployment. The system serves more than 1,000 clients, improved the internal relative bid-award KPI by 30%, and reduced inference time from 136 seconds to 25 seconds.
- **Document AI OCR** | Collected HWP/HWPX and PDF notices, parsed them with rhwp, hwp5, and liteparse v2, and normalized the output to JSON. Used Gemini 2.5 Flash-Lite only to copy target fields while deterministic code validated amounts, ratios, and units. The pipeline processes about 2,000 new notices per day and achieved F1 0.985 on key-field extraction.

Hanmac Group | ML Engineer / Researcher

Nov 2024–Feb 2026

Procurement NLP, multimodal distribution forecasting, and RAG agents

- **Multimodal PQ Bid Prediction** | Divided PQ competition into eight contexts and modeled multimodal bid distributions with R2CCP. Corrected interval collapse in the public implementation with entropy regularization and per-bin conformal thresholds, preserving separated valid ranges before estimating bid probability with 500K Monte Carlo runs. Eight models trained on 69,934 cases reached 90.73% coverage on 13,984 time-ordered validation cases and improved the group-company PQ bid KPI by 35%.
- **Procurement NLP API** | Trained a RoBERTa-large+LoRA binary model for bid eligibility, then generated detailed-domain labels through a UMAP/HDBSCAN ontology and SBERT-RoBERTa teacher-based weak labeling. Quantized the binary and multiclass models to about 330 MB with static INT8 ONNX and deployed them through a FastAPI CPU batch path. The system reached Macro-F1 0.9639 and Accuracy 96.4% for eligibility, F1 0.90 for detailed domains, and about 50 ms inference per notice.
- **Cost-Estimation RAG Agent** | Built a FAISS-based vector index from government-document OCR output, standard-cost references, and unit-price data. Connected LangChain retrieval and a LangGraph workflow to cost estimation, quantity-takeoff sheets, and official-letter drafts.

Sanha General Technology | AI / Automation Engineer

Jan 2021–Nov 2024

Document, CAD, ERP, and drone-survey automation for construction engineering

- **Document Automation** | Structured heterogeneous construction documents with OCR and layout parsing, then added confidence-based review and rule-based post-processing.
- **CAD & ERP Automation** | Built an AutoCAD add-in and template automation for repetitive drawing work and connected document and drawing outputs to ERP input workflows.
- **Drone & Survey 3D Workflow** | Converted drone imagery and survey data into 3D outputs for field review.
- **Result** | Reduced document work from more than four hours to about 30 minutes and converted outsourced or manual document and drawing tasks into in-house automation tools.

Nielsen Korea GTC | Data Analyst

Aug 2019–Oct 2020

Computer vision and RPA for global retail product data

- **Retail Vision** | Extracted operational visual attributes from global retail product images and validated data quality.
- **Barcode Data** | Connected barcode-recognition results to the product database and built a unified image and product-information workflow.

- **RPA & Exception Routing** | Automated repetitive input and validation steps, routed only exceptions for review, and reduced department-wide processing time by about 40%.

EDUCATION

Chonnam National University | B.B.A.

Mar 2015–Jul 2018

PROJECTS & RESEARCH

WORK PROJECTS

Uncertainty-Aware Bid ML

Headline q1000 and Monte Carlo decision ML for bid probability and uncertainty

Stack Python · Airflow · DAG · TabICLv2 · CatBoost · XGBoost · AutoGluon · CQR · R2CCP · Monte Carlo Simulation · GPU Inference

- **Why · How** | A single point-prediction model could not provide both bid probability and uncertainty across different competitive environments. Modeled regional competitor bids with XGBoost+CQR and R2CCP, then evaluated q1000 model-zoo candidates with Monte Carlo simulation.
- **Result** | Improved the bid KPI by 30% over the previous single-quantile model, reduced inference from 136 seconds to 25 seconds, and delivered client-level risk and bid-probability reports.

Document AI OCR

Headline HWP/PDF procurement-document extraction · F1 0.985 · 2,000 documents/day

Stack Python · HWP/HWPX · PDF · ZIP · rhwp · hwp5 · liteparse v2 · Gemini 2.5 Flash-Lite · JSON · ThreadPool

- **Why · How** | Core features such as ownership shares and capability criteria existed only in notice bodies. Collected HWP/HWPX and PDFs, normalized them to JSON, and separated LLM field copying from deterministic validation of amounts, ratios, and units.
- **Result** | Automatically processed about 2,000 notices per day and reached F1 0.985 on key-field extraction.

Procurement NLP

Headline CPU NLP API that classifies bid eligibility and detailed work domains from titles

Stack Python · PyTorch · KLUE RoBERTa-large · LoRA/PEFT · SBERT · UMAP · HDBSCAN · ONNX Runtime INT8 · FastAPI

github.com/HarimxChoi/nlp-analysis-agent

- **Why · How** | Needed immediate decisions from titles despite missing fine-grained labels and no inference GPU. Combined RoBERTa-large+LoRA, UMAP/HDBSCAN ontology construction, and SBERT-RoBERTa weak labeling, then deployed INT8 ONNX models through a FastAPI batch API.
- **Result** | The bid-eligibility model achieved Macro-F1 0.9639; detailed-domain classification achieved F1 0.90; both ran at about 50 ms per notice on CPU.

R2CCP Bid Prediction

Headline Multimodal PQ bid distributions across eight competitive contexts

Stack Python · PyTorch · R2CCP · Per-bin Conformal Threshold · Entropy Regularization · Monte Carlo Simulation

github.com/HarimxChoi/ensemble-bid-prediction

- **Why · How** | PQ scoring creates separated valid bid ranges and multimodal bid behavior. Modeled eight competitive contexts with R2CCP, then prevented interval collapse using entropy regularization and per-bin conformal thresholds before estimating bid probability with 500K Monte Carlo runs.
- **Result** | Trained eight context models on 69,934 cases, reached 90.73% coverage on 13,984 time-ordered validation cases, and improved group-company PQ bid KPI by 35% in operation.

MODEL EFFICIENCY & AGENT SYSTEMS

WarpQuant

Headline INT3 LLM quantization across weights, KV cache, and activations

Stack Python · PyTorch · Hadamard Rotation · Block-GPTQ · Output-Fisher · TurboQuant KV Cache · GGUF

github.com/HarimxChoi/WarpQuant

- **Why · How** | TurboQuant showed that Hadamard rotation could suppress KV-cache outliers. Applied the same idea to weights with block-GPTQ and Output-Fisher restoration, then added TurboQuant KV cache and token-wise activation quantization.
- **Result** | Compressed the Qwen3.8-27B text backbone to 3.6165 bpw and 11.32 GiB. It retained about 99% of BF16 performance on MMLU and common-sense evaluations; ARC rose from 52.1% to 56.8%, while PPL increased by 7.5%.

google-surf-mcp

Headline Integrated MCP for agent search, web extraction, and academic PDF parsing

Stack TypeScript · MCP · Google Search · Google Scholar · Playwright · Readability · Spatial PDF Parsing · SSRF Protection · Vitest

github.com/HarimxChoi/google-surf-mcp

- **Why · How** | Agents needed reliable access to web pages and academic PDFs beyond search snippets. Unified Google Search, Google Scholar, parallel search, Playwright/Readability extraction, spatial PDF parsing, provider fallbacks, and CAPTCHA handoff in one MCP.
- **Result** | Enabled agents to handle general search, academic search, web extraction, and PDF parsing through one MCP. Published the npm package after 388 search, extraction, recovery, and security tests; it reached the MCP TOPLIST Top 1%.

Monogram

Headline Personal knowledge management from one share action to structured, searchable knowledge

Stack Python · 5-stage LLM Pipeline · Atomic Git Tree · Markdown · Telegram · GitHub API · MCP · Dashboard

github.com/HarimxChoi/monogram

- **Why · How** | Bookmarks across news, social media, arXiv, YouTube, and reports lost context and became difficult to retrieve. Unified links, messages, documents, and code in one input path; a five-stage LLM pipeline structures content, topics, sources, and related knowledge as Markdown, while atomic Git commits preserve provenance.
- **Result** | Implemented a share-to-knowledge workflow with searchable dashboard access, versioned sources, and 13 MCP tools that reuse the same knowledge as agent context.

Bau Browser

Headline Agentic browser with user-controlled privacy and execution authority

Stack TypeScript · Electron · MCP · SQLite · Qwen3.5-4B · XGrammar · TaskSpec · ActionScope · DecisionLog

- **Why · How** | Built a local-first Electron browser that keeps browsing data, agent memory, and execution records locally. Agents propose work first; they create arguments only within approved sites and action scopes, and state-changing actions require approval and produce a receipt and DecisionLog.
- **Result** | Built a desktop MVP with native privacy controls, action approval, and execution logs. Every state-changing action follows proposal, approval, execution, and receipt; the system generated the exact action and arguments for 36/36 new-domain tasks, with zero out-of-scope actions.

LangGraph Travel Agent

Headline LangGraph travel agent that produces three bookable packages from a natural-language request

Stack Python · LangGraph · FastAPI · AsyncIO · InMemorySaver · Amadeus API · Hotelbeds API · HubSpot · Twilio

github.com/HarimxChoi/langgraph-travel-agent

- **Why · How** | For a US travel agency, structured origin, destination, dates, budget, and preferences; concurrently gathered flights, hotels, places, and restaurants; then assembled Budget, Balanced, and Premium packages with consultant review and CRM follow-up.
- **Result** | Implemented the workflow from request intake through package generation, consultant review, CRM storage, and customer contact. Delivered to a US travel agency; the IP was later transferred under contract and released as public OSS.

RESEARCH & DECISION SYSTEMS

Weakly Supervised Semantic Segmentation

Headline Commissioned WSSS SOTA research · COCO-Val mIoU 53.31% · +1.5pp vs. WeCLIP+

Stack Python · PyTorch · CLIP ViT-B/16 · DINOv2 · WeCLIP+ · RFM · COCO-Val

github.com/HarimxChoi/wsss-refined-pseudolabels

- **Why · How** | In a commissioned research project, the client asked for a WSSS method that improved on the then-SOTA WeCLIP+. We kept CLIP and DINOv2 frozen, trained only the fusion head and decoder, repaired pixels where their representations disagreed, and reused the refined pseudo-masks for self-training.
- **Result** | Achieved mIoU 53.31% on all 40,137 COCO-Val 2014 images, 1.5 percentage points above WeCLIP+ ViT-B/16 at 51.8%.

Vargo

Headline Can the best LLM-agent configuration be predicted before execution?

Stack Python · ALFWorld · 134×29 Evaluation Matrix · Router · Bandit · Activation Probe · Verifier · Sealed-cell Evaluation

- **Why · How** | Ran the same 134 ALFWorld tasks across 29 agent configurations and compared routers based on task text, embeddings, hidden states, prior results, and bandit signals. Then compared pre-execution selection with executing several candidates and verifying outcomes.
- **Result** | Pre-execution signals could not reliably identify each task's best configuration. Verified execution increased recovered oracle gap from 54% to 85% and recovered 100% within an 11-configuration deployable pool. The resulting paper is under review at TMLR.

EMH Agent

Headline Cost-aware reinforcement-learning investment agent using point-in-time data

Stack Python · TQC · IVV/IEF/SHV · Point-in-time Panel · Information Agent · Transaction Costs · Drawdown

github.com/HarimxChoi/emh-agent

- **Why · How** | Separated real-time information collection from decision making so the TQC policy used only information available at each date. Trained a three-seed ensemble to allocate IVV, IEF, and SHV, then evaluated it after transaction costs, slippage, and borrowing costs.
- **Result** | Across 756 trading days from 2019 to 2021, the cost-aware backtest returned 183.9% versus 99.6% for IVV. Maximum drawdown was -14.2% versus -33.8% for IVV.

MyShot

Headline 3D golf vision from golfer, club, and ball tracking to personalized shot prediction

Stack Python · PyTorch · RTMPose · MotionAGFormer · YOLO11 · GolfPose · GVHMR · OpenCV · Vicon · Qwen2.5-VL

- **Why · How |** Golfers need more than a pose overlay: they need to understand how joint motion and club movement affect ball speed, direction, and carry. Tracked the golfer, club, and ball, lifted RTMPose joints to 3D with MotionAGFormer, derived swing mechanics and clubhead speed, then connected physique and per-club records to a physics-based ball-flight model with pose-reliability scoring.
- **Result |** Recorded 35.6 mm MPJPE, 3.1° X-Factor MAE, and 6.1% median clubhead-speed error on GolfPose Vicon evaluation. Built 25 carry ground-truth pairs from 50 AIHub clips and redirected the system from weak body-only carry regression toward ball and club tracking with per-club calibration.

E-AT

Headline ISIC skin-lesion classification · minimum ECE 1.03% vs. CR-SAM 1.7% (AAAI 2024)

Stack Python · PyTorch · ConvNeXtV2-Tiny · Focal Loss · R-Drop · FGM · ISIC 2019 · Reliability Diagram

- **Why · How |** Combined focal loss, R-Drop, and FGM with ConvNeXtV2-Tiny on the balanced four-class ISIC 2019 dataset, tracking Macro-F1 and 15-bin ECE together by epoch.
- **Result |** Reached peak validation Macro-F1 0.8647. The minimum-ECE checkpoint recorded 1.03%, 0.67 percentage points below the then-SOTA CR-SAM's 1.7% (AAAI 2024), then evaluated classification and confidence calibration within one training framework.